



**2021**

**Preliminary  
Assessment  
Task 2**

# Preliminary Mathematics Extension

## General Instructions

- Reading Time – 2 minutes
- Working Time – 55 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 5 – 7, show relevant mathematical reasoning and/or calculations

**Total marks – 43**

**Section I – 4 marks** (pages 2 – 4)

- Attempt Questions 1 – 4
- Allow about 7 minutes for this section

**Section II – 39 marks** (pages 5 – 7)

- Attempt Questions 5 – 7
- Allow about 48 minutes for this section

**NAME:** \_\_\_\_\_

**TEACHER:** \_\_\_\_\_

| Question                 | 1 – 4 | 5   | 6   | 7   | Total |
|--------------------------|-------|-----|-----|-----|-------|
| <b>Further Algebra</b>   | /1    | /5  | /2  | /3  | /11   |
| <b>Further Functions</b> | /1    |     | /8  | /4  | /13   |
| <b>Further Trig</b>      | /2    | /9  | /4  | /4  | /19   |
|                          | /4    | /14 | /14 | /11 | /43   |

## Section I

4 marks

Attempt Questions 1- 4

Allow about 7 minutes for this section

Use the multiple-choice answer sheet for Questions 1-4

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1 What is the solution of the inequation  $\frac{4}{|x+2|} < 2$  ?

A.  $(-4, -2) \cup (-2, 0)$

B.  $[-\infty, -4, ) \cup (0, \infty]$

C.  $[-4, -2) \cup (-2, 0]$

D.  $(-\infty, -4, ) \cup (0, \infty)$

2 If  $f(x) = \frac{x}{x-2}$ , which of the following is the range of  $f^{-1}(x)$  ?

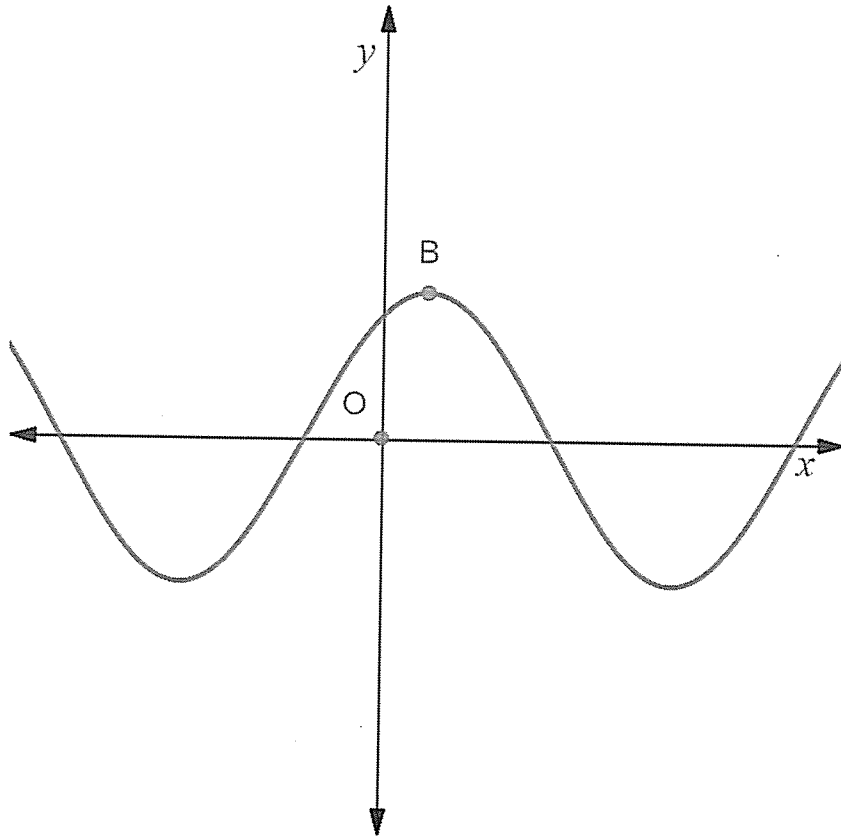
A.  $y \leq 2$

B. All real  $y, y \neq 0$

C. All real  $y, y \neq 2$

D.  $y \geq 2$

- 3 A part of the graph of the function  $f(x) = 2 \sin x + 3 \cos x$  is drawn below, but only the origin and the point  $B$ , a maximum turning point, have been labelled.



What is the  $y$  value of the point  $B$ ?

- A. 5
- B.  $\sqrt{13}$
- C.  $\sqrt{7}$
- D. 13

4 Given that  $t = \tan \frac{\theta}{2}$ , which of the following is an expression for  $\operatorname{cosec} 2\theta$ ?

A.  $\frac{(1+t^2)^2}{t^4-6t^2+1}$

B.  $\frac{t^4-6t^2+1}{(1+t^2)^2}$

C.  $\frac{(1+t^2)^2}{4t(1-t^2)}$

D.  $\frac{4t(1-t^2)}{(1+t^2)^2}$

## Section II

39 marks

Attempt Questions 5-7

Allow about 48 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 5-7, your responses should include relevant mathematical reasoning and/or calculations.

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**Question 5** (14 marks) Use a SEPARATE writing booklet.

(a) Solve  $\frac{2x}{x+2} \leq 4$ . 3

(b) Solve  $|2x - 3| < 5$ . 2

(c) By using a compound angle result, show that the exact value of  $\sin 105^\circ$  2  
is  $\frac{\sqrt{6} + \sqrt{2}}{4}$ .

(d) Let  $\sqrt{3} \cos x - \sin x \equiv R \cos(x + \alpha)$ , where  $R > 0$  and  $0 \leq \alpha \leq \frac{\pi}{2}$ .

(i) By expanding the right-hand side and equating coefficients of  $\sin x$  and  $\cos x$ , 2  
find the values of  $R$  and  $\alpha$ .

(ii) Hence, solve  $\sqrt{3} \cos x - \sin x = 0$  for  $0 \leq x \leq 2\pi$ . 2

(e) Solve  $\sin 2x - \cos x = 0$  for  $0 \leq x \leq 2\pi$ . 3

**End of Question 5**

**Question 6** (14 marks) Use a SEPARATE writing booklet.

- (a) Solve  $x(x-2)^2(x+1) \geq 0$ . **2**
- (b) Consider the function  $f(x) = \sqrt{4-x}$ .
- (i) Find  $f^{-1}(x)$  and state its domain and range. **3**
- (ii) Hence solve  $f(x) = f^{-1}(x)$ . **2**
- (c) A relation  $R$  has parametric equations  $x = 3 + 5 \cos \theta$  and  $y = 2 + 5 \sin \theta$ .
- (i) Find the Cartesian equation of  $R$ . **2**
- (ii) Describe geometrically the relation  $R$ , mentioning any important features. **1**
- (d) By using the substitution  $t = \tan \frac{x}{2}$ , solve  $\sin x + \cos x = -1$  for  $0^\circ \leq x \leq 360^\circ$ . **4**

**End of Question 6**

**Question 7** (11 marks) Use a SEPARATE writing booklet.

- (a) (i) Square both sides of the equation  $\sqrt{x} - \sqrt{y} = \sqrt{7 - \sqrt{40}}$  leaving all terms involving  $x$  and  $y$  on the left hand side. 1
- (ii) Hence find the values of  $x$  and  $y$  in the equation  $\sqrt{x} - \sqrt{y} = \sqrt{7 - \sqrt{40}}$  given that  $x > y$ . 2
- (b) The acute angles  $\alpha$  and  $\beta$  satisfy the relationships  $7 \cot^2 \alpha + 6 \cot \alpha = 1$  and  $\tan \beta = 3$ .
- (i) Determine the value of  $\tan \alpha$ . 2
- (ii) Show that  $\tan(\alpha + \beta) = -\frac{1}{2}$ . 2
- (c) A bush walker is travelling along a path described by the parametric equations  $x = t - 2$  and  $y = -2t + 9$ .  
At the same time, a lion is strolling on a course described by the parametric equations  $x = \frac{t}{2}, y = t + 1$ , where  $t$  is the time elapsed in hours after they start.
- (i) Determine where the pathways of the lion and bush walker intersect. 2
- (ii) Do the bush walker and the lion meet up with each other. Justify your answer. 2

**End of paper**

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## YEAR 11 Multiple Choice task 2 ext 1 2021

Q1/  $\frac{4}{|x+2|} < 2, x \neq -2$

$$4 < 2|x+2|$$

$$4 < 2x+4 \quad \text{OR} \quad 4 < -2x-4$$

$$0 < x$$

$$x > 0$$

$$(0, \infty)$$

$$8 < -2x$$

$$-4 > x$$

$$x < -4$$

$$(-\infty, -4)$$

(D)

Q2/  $f(x) = \frac{x}{x-2}$  D: all real  $x, x \neq 2$

$\therefore f^{-1}(x)$  Range: all real  $y, y \neq 2$

(C)

Q3/  $R = \sqrt{3^2 + 2^2}$   
 $= \sqrt{13}$

(B)

Q4/  $\operatorname{cosec} 2\theta = \frac{1}{\sin 2\theta}$   
 $= \frac{1}{2 \sin \theta \cos \theta}$

$$2 \sin \theta \cos \theta = \frac{4+t}{1+t^2} \times \frac{1-t^2}{1+t^2}$$

(C)

$\therefore \operatorname{cosec} 2\theta = \frac{(1+t^2)^2}{4+(1-t^2)}$

## 2021 Ex+1 Y-11 task 2

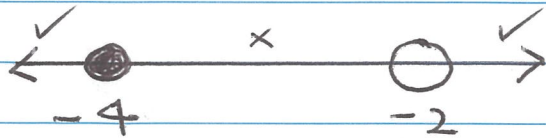
Q5)(a)  $\frac{2x}{x+2} \leq 4, x \neq -2$

Check equality

$$2x = 4x + 8$$

$$-8 = 2x$$

$$x = -4$$



$$\therefore x > -2 \text{ OR } x \leq -4$$

$$\text{OR } (-\infty, -4] \cup (-2, \infty)$$

(b)  $|2x - 3| < 5$

$$2x - 3 < 5 \quad \text{OR} \quad -2x + 3 < 5$$

$$2x < 8$$

$$-2x < 2$$

$$x < 4$$

OR

$$x > -1$$

alternatively

$$(-1, 4)$$

(c)  $\sin 105^\circ = \sin(60 + 45)$

$$= \sin 60 \cos 45 + \cos 60 \sin 45$$

$$= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{3} + 1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{6} + \sqrt{2}}{4}$$

$$(d)(i) \sqrt{3} \cos x - \sin x \equiv R \cos x \cos \alpha - R \sin x \sin \alpha$$

$$\therefore R \cos \alpha = \sqrt{3} \quad \text{and} \quad R \sin \alpha = 1$$

$$\therefore R^2 \cos^2 \alpha = 3 \quad R^2 \sin^2 \alpha = 1$$

$$R^2 \sin^2 \alpha + R^2 \cos^2 \alpha = 1 + 3$$

$$R^2 (\sin^2 \alpha + \cos^2 \alpha) = 4$$

$$R^2 = 4$$

$$R = 2$$

$$\frac{R \sin \alpha}{R \cos \alpha} = \frac{1}{\sqrt{3}}$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$(ii) \quad 2 \cos \left( x + \frac{\pi}{6} \right) = 0 \quad \frac{\pi}{6} \leq x + \frac{\pi}{6} \leq 2\pi + \frac{\pi}{6}$$

$$\cos \left( x + \frac{\pi}{6} \right) = 0$$

$$x + \frac{\pi}{6} = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$x = \frac{\pi}{3}, \frac{4\pi}{3}$$

$$(e) \quad 2 \sin x \cos x - \cos x = 0, \quad 0 \leq x \leq 2\pi$$

$$\cos x (2 \sin x - 1) = 0$$

$$\cos x = 0 \quad \text{OR} \quad \sin x = \frac{1}{2}$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}$$

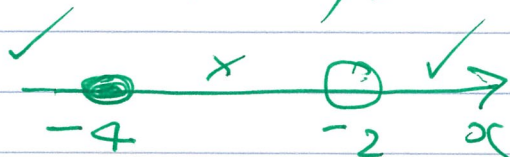
$$\therefore x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$$

$$Q1/ \frac{2x}{x+2} \leq 4 \quad x \neq -2$$

$$2x = 4x + 8 \quad (\text{solve equality})$$

$$-8 = 2x$$

$$-4 = x$$



$$x > -2 \text{ OR } x \leq -4$$

(D)

$$Q2/ f(x) = \frac{2x-1}{x+2}$$

$$\text{For } f^{-1}(x) \quad x = \frac{2y-1}{y+2}$$

$$xy + 2x = 2y - 1$$

$$2x + 1 = 2y - xy$$

$$2x + 1 = y(2 - x)$$

$$\frac{2x+1}{2-x} = y$$

(D)

$$\therefore f^{-1}(x) = \frac{2x+1}{2-x}$$

$$Q3/ \operatorname{cosec} 2\theta = \frac{1}{2 \sin \theta \cos \theta}$$

$$= \frac{1}{\frac{2(2+t)}{1+t^2} \times \frac{(1-t^2)}{1+t^2}}$$

$$= \frac{(1+t^2)^2}{4+(1-t^2)}$$

(C)

4/. Period is  $\pi$

$\therefore$  one to one is  $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] (C)$

Q5/(a)  $x^2 - 3x + 2 \geq 0$   
 $(x-2)(x-1) \geq 0$



$\therefore x \geq 2$  OR  $x \leq 1$

(b)  $\sqrt{x+5} = 7-x$ ,  $x \geq -5$

$$x+5 = 49 - 14x + x^2$$

$$0 = x^2 - 15x - 44$$

$$0 = (x-4)(x-11)$$

$$x = 4 \text{ OR } x = 11$$

check

$$\sqrt{9} = 7-4$$

$$3 = 3$$

TRUE

check

$$\sqrt{11+5} = 7-11$$

$$4 \neq -4$$

not a solution

$\therefore x = 4$

(c)  $3^{2x-3} \cdot 2^{2x-3} \cdot 2^{-2x+2} = 3^{\frac{1}{2}} \cdot 2^{-1}$

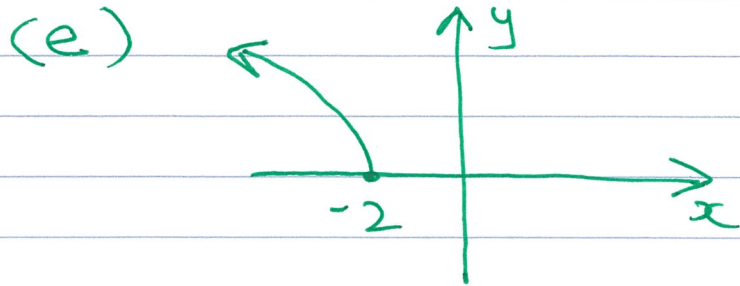
$$3^{2x-3} \cdot 2^{\cancel{-1}} = 3^{\frac{1}{2}} \cdot 2^{\cancel{-1}}$$

$$2x-3 = \frac{1}{2}$$

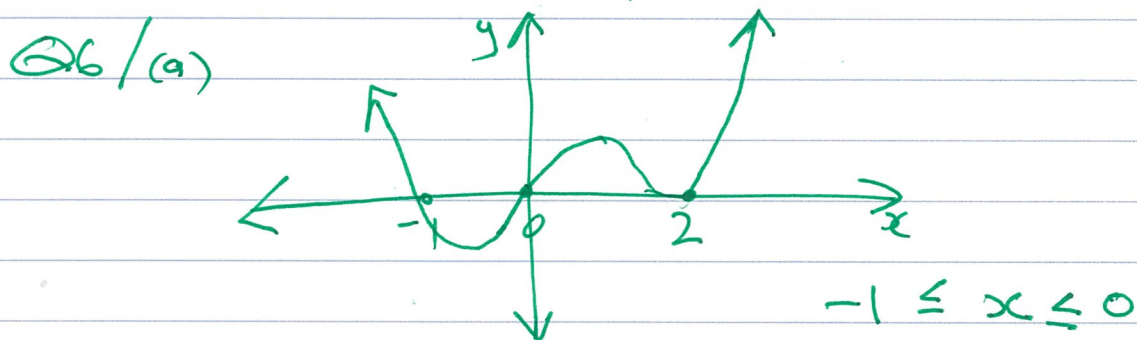
$$2x = \frac{7}{2}$$

$$x = \frac{7}{4}$$

$$(d) (-\infty, 2) \cup (2, \infty)$$



$$\begin{aligned}
 (f) \quad & 4 (\cos x \sin x)^2 \\
 &= 4 \left( \frac{1}{2} \sin 2x \right)^2 \\
 &= 4 \times \frac{1}{4} \sin^2 2x \\
 &= \sin^2 2x
 \end{aligned}$$



$$(b) \quad \sqrt{x} - \sqrt{y} = \sqrt{7 - \sqrt{40}}$$

$$\begin{aligned}
 x - 2\sqrt{xy} + y &= 7 - \sqrt{40} \\
 &= 7 - 2\sqrt{10}
 \end{aligned}$$

$$\therefore x + y = 7 \quad \text{and} \quad xy = 10$$

$$\therefore x = 5, y = 2 \text{ by inspection or solving.}$$

$$(c) \frac{x-3}{5} = \cos \theta \quad \frac{y-2}{5} = \sin \theta$$

$$(i) \frac{(x-3)^2}{25} + \frac{(y-2)^2}{25} = \sin^2 \theta + \cos^2 \theta$$

$$\therefore (x-3)^2 + (y-2)^2 = 25$$

(ii) Circle centre (3, 2) radius = 5

$$(d) \cos 105 = \cos 60 \cos 45 - \sin 60 \sin 45$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{1 - \sqrt{3}}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{2} - \sqrt{6}}{4}$$

$$(e) f(x) = 2 \sin\left(\frac{\pi}{2} - 3x\right)$$

$$= 2 \cos 3x$$

$$f(-x) = 2 \cos(-3x)$$

$$= 2 \cos 3x$$

$\therefore f(x) = f(-x)$  So it's an even function

$$\begin{aligned}
 (f) \quad \sin[(a+b)-(a-b)] \\
 &= \sin[a+b-a+b] \\
 &= \sin 2b
 \end{aligned}$$

Q7(a)(i)  $x+2=t$        $y=-2(x+2)+9$   
 $y=-2x-4+9$   
 $y=-2x+5$  (Cartesian equation of bushwalker)

$t=2x$        $y=2x+1$   
 (Cartesian equation of lion)

$$2x+1=-2x+5$$

$$4x=4$$

$$x=1$$

$$y=3 \quad \therefore \text{paths meet at } (1,3)$$

(ii) When bushwalker is at  $(1,3)$   $t=3$   
 When lion is at  $(1,3)$   $t=2$

$\therefore$  They don't meet.

$$(b) \sqrt{3} \cos x - \sin x \equiv R \cos x \cos \alpha + R \sin x \sin \alpha$$

$$\therefore \left. \begin{aligned} R \sin \alpha &= 1 \\ R \cos \alpha &= \sqrt{3} \end{aligned} \right\} \text{Divide}$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$\left. \begin{aligned} R^2 \sin^2 \alpha &= 1 \\ R^2 \cos^2 \alpha &= 3 \end{aligned} \right\} \text{Add}$$

$$R^2 = 4$$

$$R = 2 \text{ (as } R > 0 \text{)}$$

$$\therefore \sqrt{3} \cos x - \sin x = 2 \cos \left( x + \frac{\pi}{6} \right)$$

(ii) As  $R = 2$  minimum value is  $-2$

$$\therefore 2 \cos \left( x + \frac{\pi}{6} \right) = -2$$

$$\cos \left( x + \frac{\pi}{6} \right) = -1$$

$$x + \frac{\pi}{6} = \pi$$

$$x = \pi - \frac{\pi}{6}$$

$$x = \frac{5\pi}{6}$$

$$\begin{aligned} (c)(i) x^3 - x^2 + x - 1 &= x^2(x-1) + 1(x-1) \\ &= (x-1)(x^2+1) \end{aligned}$$

$$(ii) \text{ LHS} = \frac{\frac{1}{\sin \theta} - \sin \theta}{\frac{1}{\cos \theta} - \cos \theta}$$

$$= \frac{1 - \sin^2 \theta}{\sin \theta} \div \frac{1 - \cos^2 \theta}{\cos \theta}$$

$$= \frac{\cos^2 \theta}{\sin \theta} \times \frac{\cos \theta}{\sin^2 \theta}$$

$$= \frac{\cos^3 \theta}{\sin^3 \theta}$$

$$= \cot^3 \theta$$

$$(iii) \quad \cot^3 \theta + \cot \theta = 1 + \cot^2 \theta$$

$$\cot^3 \theta - \cot^2 \theta + \cot \theta - 1 = 0$$

Using the part (i)

$$\cot \theta = x$$

$$(\cot^2 \theta + 1)(\cot \theta - 1) = 0$$

$$\cot^2 \theta = -1$$

no solutions

$$\cot \theta = 1$$

$$\therefore \tan \theta = 1$$

$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}$$